

MOHAMMED VASHIR

CREATIVE TECHNOLOGIST · SYSTEMS ARCHITECT · PHYSICAL COMPUTING

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EXECUTIVE PROFILE

Accomplished Systems Engineer and Creative Technologist specializing in physical computing, autonomous robotics, multilayer RF hardware design, and high-performance 3D spatial web applications. Founder of 4tech Engineering R&D with a proven track record of independently designing, fabricating, and empirically validating 15 production-grade hardware systems. Proficient at bridging bare-metal embedded firmware (C++20, FreeRTOS) with precision kinematic control and real-time Three.js/WebGL digital twin visualization.

CORE TECHNICAL SPECIALIZATIONS

- **Robotics & Kinematics:** Analytical Closed-Form Inverse Kinematics, Denavit-Hartenberg frames, Harmonic Drive actuators, precision laser focal tracking, robotic gripper physics.
- **High-Speed PCB Design:** 6-Layer controlled impedance routing, 50:1 ΔZ differential pairs, KiCad, Altium, EMI/EMC shielding, high-current motor drivers.
- **Embedded Firmware & IoT:** Bare-metal C++20, FreeRTOS, STM32H7 Dual-Core ARM Cortex-M @ 480MHz, ESP32, microsecond ISR latency, hardware AES-256 encryption, zero heap allocations.
- **Autonomous Systems & RF:** ROS2 Humble nodes, 2D/3D LiDAR SLAM, 15-state EKF3 sensor fusion, Sub-GHz (433/868/915MHz) telemetry nodes (-142dBm LNA), polar radiation scanners.
- **Spatial Web & Graphics:** Three.js, WebGL/WebGPU, custom GLSL vertex/fragment shaders, ACESFilmic tonemapping, Web Audio API synthesis, React, Vite.
- **Fabrication & Testing:** CNC machining, 3D printing (FDM/SLA), logic analyzers, GHz oscilloscopes, SDR spectrum analyzers, SolidWorks CAD modeling.

ENGINEERING LEADERSHIP & EXPERIENCE

4tech Engineering R&D Atelier — *Founder & Principal Systems Architect*

2022 – PRESENT | KALPAKKAM, INDIA

- Spearheaded research, schematic design, high-speed PCB layout, structural fabrication, and firmware development across 15 production-grade hardware systems.
- Formulated closed-form trigonometric Inverse Kinematics solvers running in real time on microcontrollers for 6-axis articulated manipulators with zero singular matrix jitter.
- Developed an autonomous flight computer running a 15-state Extended Kalman Filter (EKF3) with triple IMU redundancy, sub-GHz telemetry dispatch, and GPS-denied obstacle clearance.
- Constructed automated radio frequency testbenches capable of continuous 360-degree antenna radiation pattern profiling and polar plot synthesis.
- Engineered museum-grade interactive 3D WebGL portfolios and digital twins connecting physical sensors directly to browser canvases via low-latency WebSockets.

SELECTED FLAGSHIP ENGINEERING SYSTEMS

6-DOF Articulated Robotic Arm (Robotics & Kinematics)

Titanium and twin-spar aluminum manipulator featuring harmonic drive actuators, optical rotary encoders, real-time closed-form analytical IK, parallel gripper, and cursor-aligned optical targeting laser.

Precision Dual-Core UAV Autopilot (Aerospace & Autonomous Control)

Autonomous flight management computer powered by STM32H7 dual-core ARM @ 480MHz, 15-state EKF3 navigation filter, triple redundant barometric & IMU sensors, and 868MHz spread-spectrum datalink.

Autonomous SLAM Inspection Rover (Autonomous Ground Vehicles)

All-terrain 4WD planetary drivetrain rover equipped with 360° LiDAR, stereo depth vision, and ROS2 Humble nodes for GPS-denied indoor cartography and structural facility inspection.

Sub-GHz Long-Range Telemetry Node (RF Telecommunications)

Ultra-low power (1.2 μ A standby) industrial telemetry node operating across 433/868/915MHz with -142dBm sensitivity, LNA front-end, hardware AES-256 GCM encryption, and 24km line-of-sight range.

EDUCATION & CREDENTIALS

Bachelor of Technology (B.Tech) in Electrical & Electronics Engineering

FIRST CLASS HONORS

Coursework: Embedded Systems, Control Theory, Electromagnetic Radiation, Microcontrollers, Signals & Systems